Project Number: 58474-001

Armenia: Empowering Mobility through Electric Urban Solutions (EMERGE)

Terms of Reference

Conceptual Design and Due Diligence for the Yerevan Sustainable Urban Transport and Electric Mobility Project

April 2025

ABBREVIATIONS

ADB	Asian Development Bank
BRT	Bus Rapid Transit
CCTV	Closed Caption Television
CDIA	Cities Development Initiative for Asia
CS	Commencement of Services
EBRD	European Bank for Reconstruction and Development
EIA	Environmental Impact Assessment
EMP	Environmental Management Plan
FVO	Frequency and Vehicle Occupancy
GAP	Gender Action Plan
GCF	Green Climate Fund
GOA	Government of Armenia
IEE	Initial Environmental Examination
iRAP	International Road Assessment Program
ITS	Intelligent Transportation Systems
OD	Origin-Destination
LARP	Land Acquisition and Resettlement Plan
MTAI	Ministry of Territorial Administration and Infrastructure
NMT	Non-Motorized Transport
PIU	Project Implementation Unit
PKI	Passenger-Kilometer Index
PV	Photovoltaic
RFID	Radio Frequency Identification
QCBS	Quality-Cost Based Selection
SPS	Safeguard Policy Statement
SUDIP	Sustainable Urban Development Investment Program
TA	Technical Assistance
YM	Yerevan Municipality

Conceptual Design and Due Diligence for the Yerevan Sustainable Urban Transport and Electric Mobility Project

Terms of Reference

1. Project Background

1. Armenia is a landlocked country in the Caucasus region. Its capital, Yerevan, is the largest city and the financial, industrial, and commercial center. The city's gross domestic product accounts for more than 62% of Armenia's gross domestic product. With a population of about 1.2 million people, Yerevan is situated along the Hrazdan River, which divides the northern and western residential suburbs from the city center. Yerevan's basic urban services fail to meet increasing urban household and commercial demand, leading to a gradual degradation of the urban environment and living standards. The urban poverty rate in 2021 was 21.8% against a national poverty rate of 26.5%.

2. Public transport services are provided by buses, minibuses, trolleybuses, and the underground metro system, which were developed without a consolidated urban transport master plan. Public transport management and regulation are poorly coordinated, and bus and minibus services are unreliable. A lack of reliable public transport results in reliance on personal car use, further exacerbating congestion. Yerevan Municipality (YM) aims to enhance urban mobility and traffic conditions in the city through improvements of public transport services and the road network. It has started a bus fleet renovation program and continues discussions with key development partners such as the Asian Development Bank (ADB) and the European Bank for Reconstruction and Development (EBRD). ADB has been working with Yerevan Municipality through the Sustainable Urban Development Investment Program (SUDIP) to improve the urban road network and provide guidance on how to improve public transport. EBRD has supported the renovation of the bus fleet and rehabilitation of the metro system.

3. The Government of Armenia (GOA) has requested the ADB to help finance improvements of bus-based public transport, to improve the quality of and increase the demand for public transport in Yerevan and curb the increasing demand for the use of private vehicles. The project would be supported by the Green Climate Fund (GCF) through the E-Mobility Program, which includes improvements of non-motorized and public transport, and the deployment of electric buses.

4. After discussions with the ADB, the Government of Armenia, through the Ministry of Territorial Administration and Infrastructure (MTAI) and the Municipality of Yerevan, has formally requested support from the Cities Development Initiative for Asia (CDIA)[1] to prepare prefeasibility and feasibility assessments for this project.

5. The Executing Agency (EA) is the Yerevan Municipality (YM). The Yerevan Urban Development Programs Implementation Unit (PIU) will be responsible for the day-to-day project implementation.

[1] CDIA is a multi-donor trust fund managed by the Asian Development Bank (ADB). As a project preparation facility, it works with secondary cities in Asia and the Pacific to address gaps in infrastructure development and financing. It supports projects that emphasize poverty reduction, environmental improvement, climate change mitigation or adaptation, and good governance. CDIA receives funding support from Austria, Germany, the Republic of Korea, Spain and Switzerland.

2. Purpose of the Assignment

6. ADB-CDIA requires the services of a qualified international consulting firm to develop: (i) bus public transport service Plan for Yerevan; (ii) Conceptual Design of two bus priority corridors; (iii) Upgraded depots to accommodate a fleet of battery electric buses; (iv) Traffic and Parking Management Plan; (v) Communications and Branding; and (vi) due diligence for ADB's consideration of project financing.

7. Consultant selection and engagement will be carried out in accordance with ADB's Guidelines on the Use of Consultants (2017, as amended from time to time), using the quality-cost based selection (QCBS) method with a standard quality-cost ratio of 90:10.

3. Duration and Location of the Services

8. The consultancy services are required for 12 calendar months, expected to commence in May 2025. The location is Yerevan, Armenia.

4. Scope of Services

9. The Government of Armenia is working with ADB in Yerevan to develop:

a. Improved public transport services.

b. Two bus priority corridors.

c. New/upgraded bus depots to accommodate the future electric bus fleet.

10. Recent studies have provided a series of recommendations to public transport services, including bus route restructuring and priority measures, integrated tariff and ticketing system, fleet renovation, deployment of electric buses, bus depot development, institutional reform, and preliminary design of a pilot bus priority corridor.

11. The two bus priority corridors (shown below) are as follows:

- Grigor Lusavorich, Arshakunyats, Garegin Nzdeh and Bagratunyats streets (yellow below, 9 kilometers)
- Nor Nork, Arabkir, and Malatia Sebastia districts (purple below, 19 kilometers).

12. The total length of these corridors is about 28 kilometers.

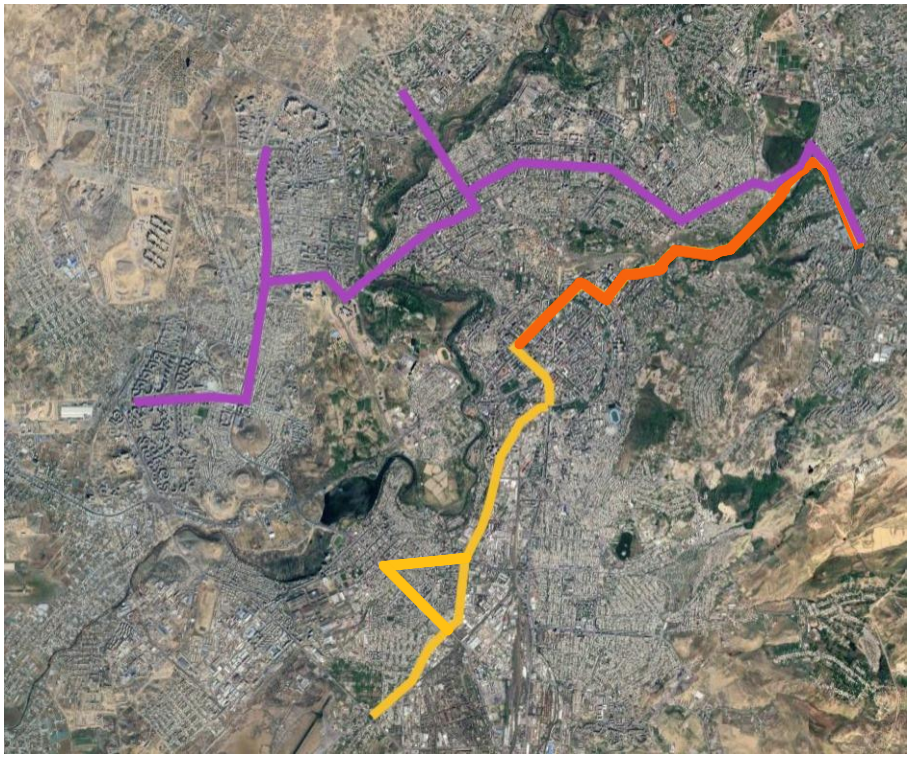


Figure 1: Two bus priority corridors for Yerevan, totaling about 28 kilometers

13. These corridors will be finalized during the inception stage of the project; however, during the course of the project, the need for small changes may be identified.

14. The referred studies, which will be provided to the Consultant, include:

(i) 2019: New Bus Network and Integrated Tariff and Ticketing System, WYG International Limited in partnership with Ameria CJSC
(ii) 2020: Yerevan Bus Depot Renovation Plan and Network Overview, BRT Planning International
(iii) 2020: E-Bus Options for Yerevan, Grütter Consulting
(iv) 2022: Project Preparation Study for the Yerevan Sustainable Urban Transport Implementation Project, Mott MacDonald Limited
(v) 2024: Yerevan Public Transport Improvements, Ile-de-France Mobilites

15. The scope of the required services described herein includes the conceptual design of these project components and due diligence for ADB's consideration of project financing.

5. Detailed Tasks, Outputs and Expected Deliverables of the Assignment

16. The required activities and outputs are summarized in Table 1 below.

Table 1: Key Activities and Outputs

Task 1 Service Plan for the Bus Priority Corridors	
1.1	Data Collection
1.2	Bus Priority Service Plan
Task 2 Conceptual Design for Two Bus Priority Corridors	
2.1	Full Conceptual Designs and Costings for Bus Priority Corridors
2.2	3-D Renderings
Task 3 Traffic and Parking Analysis	
3.1	Traffic Surveys and Analysis
3.2	Traffic Management Plan
3.3	On-Street Parking Plan
Task 4 Conceptual Design and Upgrading for 2 Bus Depots	
4.1	Conceptual design of Bagratunyats Depot
4.2	Conceptual design of Jrvezh Depot
Task 5 Communications and Branding	
5.1	Stakeholder outreach during the conceptual design phase
5.2	Branding
5.3	Stakeholder plan during project implementation
Task 6 Safeguards	
6.1	Initial Environmental Examination (IEE) or Environmental Impact Assessment (EIA), including environmental management plan (EMP)
6.2	Land Acquisition and Resettlement Plan (LARP)
6.3	Ethnic Groups Due Diligence Report
Task 7 Economic and Financial Analysis of the Project	
7.1	Financial Management Assessment
7.2	Financial Analysis
7.3	Economic Analysis
Task 8 Procurement Planning	
8.1	Strategic Procurement Plan (SPP)
8.2	Procurement Support
Task 9 Climate Change Assessment	
9.1	Climate Risk and Adaptation Assessment
9.2	Climate Change Assessment
Task 10 Poverty, Social and Gender	
10.1	Poverty, Social and Gender Assessment
10.2	Social Development and Gender Action Plan
Task 11 Institutional Assessment and Capacity Building	
11.1	Institutional Assessment
11.2	Institutional and Capacity Building Roadmap
11.3	Report on Knowledge Transfer Activities

17. The following tasks must be completed by the consultant to deliver the outputs listed above.

Task 1: Service Plan for the Bus Priority Corridors

18. The bus priority corridors for Yerevan are envisioned "Direct Service" corridors. That means that many different bus routes will use the bus priority infrastructure at different points along the corridors, turning onto and off the corridors as needed. A Direct Service bus priority corridor reduces the number of transfers that passengers need to make in order to reach their destination.

19. In this task, the Consultant will develop a Direct Service service plan for each of the two identified bus priority corridors.

Develop bus map and determine "considered routes"

20. In order to develop a service plan, all routes which currently interface with the bus priority corridors must be identified, as these routes will form the basis of the future service plan.

(i) Confirm and map the bus routes:
The website of the Yerevan Municipality includes tables of all bus, minibus, and trolleybus routes operating in Yerevan today.[2] This includes 76 bus/minibus routes and 5 trolleybus routes.

21. Using a combination of field verification and verification using the bus tracking app Yandex, the Consultant will verify all routes in the network and place them on a GIS map.

22. Once the map has been finalized, the Consultant will identify those bus routes which either:

- Overlap either of the two bus priority corridors;
- Intersect anywhere along either of the bus priority corridors; or
- Operate in parallel to either of the bus priority corridors within 500m of any portion of either corridor.

23. These "considered routes" will then constitute the routes which will move forward for further analysis in the rest of this task. It is important at this stage to consider all of the routes which meet the above criteria as "affected" and not only those routes with significant overlap on the corridor, because the next stage of this task will include possible changes to the current routes, before a BRT service plan is finalized. The Consultant will develop a brief memo outlining which routes have been selected for further analysis, with justification based on the criteria laid out above.

Collect data for the considered routes

24. To develop an operationally efficient service plan which benefits passengers, the Consultant will collect baseline data related to the considered bus routes identified above. Some data is already available through the various studies listed above. However, the Consultant should assume that all data obtained through previous studies will need to be collected and/or verified in

2 https://www.yerevan.am/hy/route-network/

the field. Prior to the start of the field data collection, the Consultant will develop a memo describing their methodology for field data collection, to be signed off on by the Transport Department ahead of data collection activities. AM and PM peak hours should be determined by the Consultant, together with the Transport Department and included in the memo.

25. The following is a list of all data that is needed:

(i) **On-board surveys:** The Consultant will conduct on-board surveys for all considered routes. During these surveys, surveyors will record stopping locations, times, and boarding and alighting volumes at all stops. On-board surveys will be conducted for a number of round trips equal to half the peak hour frequency on each considered route, with that number divided again between AM and PM peak hours. .

(ii) **Cause of delay surveys:** The Consultant will conduct cause of delay surveys by walking the length of each corridor twice in the AM peak and twice in the PM peak, noting specific factors which are delaying buses. A high level of specificity is required for this survey, with a written description of precisely what is causing any delay which is noted.

(iii) **Frequency and vehicle occupancy surveys:** The Consultant will conduct detailed frequency and vehicle occupancy (FVO) surveys at key points along all the priority bus corridors, as well as at some key points along the considered routes. Surveys will be conducted in the peak hour, peak direction. At each location, the following data should be collected for each public transport vehicle which passes the cordon line:

- Vehicle type/size (e.g., 12m bus, 18m bus, trolleybus, etc.)
- Route number
- Approximate occupancy of the vehicle (0%, 25%, 50%, 75%, 100%)
- Time that the vehicle passed the cordon line.

FVO points should be selected together with the PIU.

(iv) **Boarding and alighting surveys:** At all existing bus stops along the bus priority corridors, the Consultant will conduct boarding and alighting surveys to cover the morning and evening peak four-hour period, in the peak direction. At each stop, surveyors should record:

- Bus route number
- Number of passengers boarding
- Number of passengers alighting
- Time (in 5- or 10-minute increments)

(v) **Transfer survey:** The Consultant shall conduct transfer surveys at least five of the highest transfer points on or near the bus priority corridors. Surveyors should collect origin-destination (OD) data by asking passengers from which route they are coming, and to which route they are transferring. Surveyors should also ask passengers about their door-to-door origin and destination. The Consultant will select the survey locations together with the Transport Department.

26. The Consultant will then create maps of all the data including:

(i) Map of full bus network and individual route maps

(ii) Maps showing aggregated link loads segment by segment, for all considered routes.

(iii) Boarding and alighting maps by route and aggregated with specific maps showing aggregated boardings and alightings along the bus priority corridor.

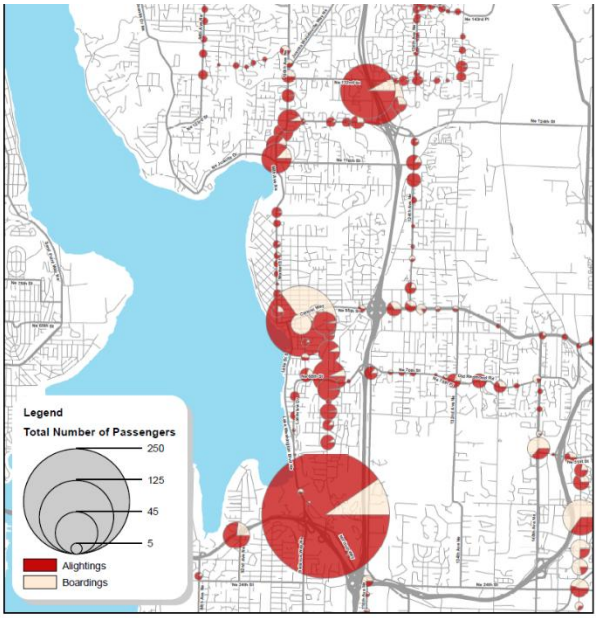


Figure 2: Boarding and alighting volumes, aggregated for all bus routes in Kirkland, Washington outbound during the PM peak.

(iv) Transfer maps showing breakdown of transfer volumes from each route passing through major transfer points and continuing onward.

(v) Bus speed maps and maps highlighting major causes of delay.

27. **Network Diagnostics:** Using the data and maps generated above, determine what is working and what is not about the existing bus network, namely the considered routes. The Consultant should specifically consider areas of slow speeds, forced transfers, new developments unserved by transit, indirectness of route, etc. The Consultant should measure the Passenger-Kilometer Index (PKI) for each route in order to determine the level of productivity of each route. The Consultant should prepare a PowerPoint, describing their diagnosis of the network and present it to the PIU and other relevant stakeholders before moving to the next stage. Network Diagnostics should also form the beginning of the Service Plan report.

28. **Service Plan:** Based on the diagnosis of problems in the existing network, as well as the government's priorities, and finally, on a review of the previous studies undertaken (paragraph 13) related to the bus network, the Consultant should propose changes, where needed, to the considered routes. The Consultant should not assume that the routes in the existing network will remain as is and either become part of the service plan or be left out. It may be possible that routes near the corridors could be rerouted onto the corridors; that routes that intersect the

corridors may be joined with routes currently using the corridors, or extended onto the corridors, etc. Through careful calculation of corridor saturation, the Consultant should determine which routes should become part of the bus priority corridor, using the best practice formulas developed for this purpose.[3]

29. The Consultant should design services from the perspective of state-of-the art service planning.[4] If some critical OD pairs are unserved, especially between low-income neighborhoods and hospitals, educational establishments, major employment areas and markets and major food retail centers, the Consultant may propose new routes. Consideration should also be given to highlighting accessibility (through mapping travel times) delivered by the existing and proposed network to key destinations such as employment areas, hospitals and health centers, education establishments and markets and food retail locations. If some critical destinations are unserved, especially by predominantly low-income neighborhoods, then the Consultant may propose new routes.

30. Each route should be presented with its new PKI indicator.

31. On the bus priority corridors, the Consultant will confirm station locations based on the demand patterns on the various routes, as well as right-of-way availability and best practices in station spacing on bus priority corridors such as BRT corridors.

32. Off the bus priority corridors, the Consultant will propose bus stop locations for the routes that will be developed above. Location will depend on:

(i) Existing bus stop locations: basing bus stop locations on the locations of existing bus stops, where proposed routes overlap new routes, will help to minimize new implementation costs, and maintain user habits.

(ii) Optimization of bus stops: Some bus stops will need to be relocated in order to optimize travel time for passengers on board versus walking time for passengers accessing the stops. Some observations at and around key destinations will be needed in order to better understand walking distances for passengers.

(iii) Access to destination points: where new lines are proposed, bus stops should be located as near as possible to destination points.

(iv) Safety: bus stops should be located to ensure pedestrian safety (road crossings, good visibility) and car safety (good visibility, ability to stop).

33. Then, demand, frequency, bus size, and fleet size on each route should be estimated.

34. The Consultant shall develop high-quality graphic-designed maps of the new network and each individual route. New and/or changed routes should be shown alongside existing routes with similar itineraries for public outreach purposes. The Consultant shall submit all graphic design files in an editable format in case routing changes at a later time.

[3] Ibid.
[4] Detailed guidance on service planning is provided in the BRT Planning Guide, 4th Edition, https:/www.itdp.org/the-brt-planning-guide/

35. General off-corridor bus stop sizing should be proposed, particularly where larger volumes of passengers and/or buses are expected.

36. **Station and stop sizing.** Based on projected boarding and alighting volumes, plus 20% for growth, the Consultant will define the sizing requirements of the stations to meet the operational characteristics, including the number of platform segments and the number of stopping bays. It is critical that this occur prior to conceptual design of the corridors (Task 2). Also, the Consultant will define the stopping bay location for each route utilizing a particular station; each route shall have a pre-defined stopping location to maximize customer convenience and service reliability.

37. The Consultant shall identify one or more staging area locations in or near the city center. The staging area locations shall be selected to minimize dead kilometers (i.e., non-revenue kilometers) and where land is available.

38. **Service Planning Workshop.** Following the development of the Service Plan, the Consultant will hold a workshop, including key stakeholders, in which the proposed network is discussed. Feedback should be collected and incorporated into the final Service Plan.

39. **Deliverables:**

(i) Memo describing selection of considered routes for further analysis

(ii) Data collection methodology memo

(iii) Raw and processed datasets for all collected data (Excel), including maps

(iv) Network diagnostics PowerPoint

(v) Service Plan PowerPoint and report, including high-quality, graphic-designed maps of the network, each route, and changes to existing routes.

Task 2: Conceptual Design for Two Bus Priority Corridors

40. In this task, the Consultant will develop conceptual-level drawings for the two bus priority corridors described above. It is critical to note that the corridor design should align with international road safety standards such as Economic Commission for Europe (ECE) bus safety regulations and should achieve full universal accessibility. The final design for road infrastructure is to meet at least a 3 Star Safety Rating using the iRAP methodology. The drawings will include, but not be limited to, the following:

a) Roadway cross-sections (bus lanes and bus lane delineators, stations, bicycle lanes, pedestrian facilities including sidewalks, crossings and signalization, landscaping, etc.)
b) Station architecture and station components (ramps, balustrades, shutters, kiosk, gate area, flooring, top structure, platform-vehicle interface, totem, customer signage, bicycle paring facilities, and public toilet facilities)

41. At each station, the Consultant should assess walkability, accessibility, and pedestrian safety for a minimum distance of 300 meters in all walkable directions. Crash data (collected in Task 3) should be used as part of this assessment. The Consultant will then prioritize roughly

50km of walkability upgrades to stations on the two corridors. At off-corridor bus stops used by routes using the bus priority corridor, the Consultant should propose required interventions following a prioritization scheme. The stations will also include tactile tiles to assist the sight impaired. Braille readers will also be available to provide system information and instructions.

42. **Site-specific surveys:** The Consultant shall collect the following data, sufficient to carry out conceptual design of the bus priority corridors:

(i) **Extent of right-of-way:** Using government records, the Consultant will confirm the extent of the existing right-of-way as a first step to developing the design. Where records are unavailable, the Consultant shall map the existing right-of-way.
(ii) **Current road geometry:** The Consultant will record and measure existing lane widths and usage
(iii) **Signal timings and phasings:** Signal timings and phasings at all intersections along the corridors should be available from the government but if not, the Consultant shall collect this information in the field.

43. **Conceptual roadway cross sections.** The Consultant will develop conceptual cross sections for each unique section of roadway. Alignment of the bus lanes should be designed to reduce the delay recorded in Task 1 as much as possible. Calculations should be made demonstrating how the alignment of the bus lanes impacts delay under both central median and curb-aligned alternatives. Bus lane delineators should be discussed with the Transport Department and included in the conceptual roadway cross sections.

44. The Consultant will also include bicycle lanes in the cross sections, ideally, extended throughout the length of the bus priority corridors.

45. Wide sidewalks should be included throughout the length of the corridors. Landscaping and other amenities should be included.

46. **Develop conceptual-level design for bus priority stations.** The Consultant will develop conceptual-level station designs for each type/size of station. Station design should follow all principles of universal accessibility and should be gender sensitive. Core design components of the stations will include, but are not limited to, the following: pedestrian access, bicycle parking, public toilets (where possible and mainly at larger, more primary stations), entry ramps and balustrades, station shutters (for shutting the stations at non-operational times), fare sales kiosk (including space for commercial sales), gate area, base structure, top structure, flooring materials, ceiling, energy-efficient lighting, air purification system, heating, universal power box, cash drop box, vehicle-platform interface, doorways, gutters, real-time information signs, totem, etc. Bike share stations should be explored, with consideration to the extent and model of the existing bike share system in Yerevan. The station platform should be designed to be at the same level as the floor of the buses, so that passengers with wheelchairs or strollers can roll directly onto and off the bus.

47. Stations and stops should be sized according first, to the station sizing calculated in Task 1 (paragraph 36) and second, to the space available. Passing lanes should be included at stations, as dictated by demand and frequency calculations, as well as by service patterns (express/local services) and as space allows.

48. Station area footprints should include integration with pedestrian access and cycleway facilities. Along the bus priority corridors, the Consultant shall design a full shelter, including weather protection, information system and signage, and seating. At least three options for the top structure architectural design shall be produced, shown in realistic 3-D renderings. The Consultant will present the two options to key stakeholders, during which time the final architectural design will be selected. For the stops on the off-corridor portions of the network (i.e. segments without dedicated busway), the Consultant will develop the design of the curbside shelters and/or stop indicators, as well as pedestrian access.

49. **Conceptual design of vehicle-platform interface.** The design of the station-to-vehicle interface should be given particular attention. The Consultant will develop the docking assist measures to ensure the platform-to-vehicle gap is minimized to the extent possible. Alignment roadway markers will be utilized to guide the vehicle as it approaches the station. Roadway channels can be used to provide rapid and accurate alignment of the tire footprints relative to the station. In addition, as practiced on the ADB-financed project in Yichang, PRC, boarding bridge extensions from the station platform can effectively eliminate the gap problem in its entirety. The Consultant is encouraged to pursue all of these techniques through a comprehensive detailed design of the vehicle-platform interface.

50. **Integration with Yerevan Metro.** The bus priority corridors will cross the Yerevan Metro at Barekamutyun and Yeritasardakan. Where the bus priority corridor intersects with the Metro, the Consultant should develop integrated stations for seamless transfers between the modes. Ideally, this will consist of a direct connection from the bus station platform down to the metro. Every attempt should be made to integrate fare payment as well so that only a single swipe is required.

51. **Conceptual staging area design.** The Consultant will develop a conceptual design for one or more vehicle staging areas in or near the inner-city area. The staging areas will allow vehicles to be parked and partially serviced without having to return to the distant depot area. The sites for the staging areas will come from the service plan (Task 1). The Consultant will evaluate each potential staging area site in conjunction with the PIU. The Consultant will also provide valuation estimates of any site in which compensation is required to purchase the property for the use as a vehicle staging area.

52. **Parking delineation.** A full analysis of parking supply, occupancy, and future needs on and around the corridor will be carried out simultaneously, under Task 3. Where necessary, some parking will be eliminated from the corridor and analysis will determine where displacement to side streets will be possible. For parking that remains both on- and within 100m off-corridor, the Consultant will design structured parking spaces. Between every 3-4 parking spaces, landscaped curb extensions shall be provided.

53. **Route turnarounds.** At the terminus of each proposed BRT route, the Consultant shall ensure that bus turnarounds are designed with minimal impacts to bus operations and mixed traffic. Dedicated facilities should be considered, including U-turns, left turns, etc.

54. **Intersections.** All intersections should include safe pedestrian crossings, with curb cuts on either side. Tabletop crossings should be considered, where possible, as well as cycle tracks that continue through intersections. Where intersections are overly complicated, causing delays to buses and other traffic, the Consultant should consider simplification. The Consultant should design bus priority measures at intersections, such as queue jumps, signal priority, or reducing or eliminating any conflict with turning vehicles. This may mean eliminating turns across the bus lanes and designing U-turns beyond intersections. Where traffic signals are proposed, the

Consultant will propose phasing. Two-phase intersections are recommended. A pedestrian walk phase must be included. The Consultant must also ensure that all turning movements for bus routes, as designed in the service plan, will be accommodated as seamlessly as possible.

55. **Utility design.** The Consultant will assess the state of the existing utilities along the corridors and will assess the government's interest in upgrading utilities, burying power lines, installing fiber optic cable, etc. The Consultant will then provide a conceptual-level utility design and costing.

56. **Intelligent Transportation Systems (ITS) design.** The Consultant should determine what types of ITS are needed for the project. This could include CCTV systems, off-board fare collection technology, operational control system, signal priority for buses, etc. To do this, an assessment of existing ITS related to public transport in Yerevan will be necessary. The Consultant will review all existing systems and determine whether changes/upgrades are needed in order to bring the bus system up to state-of-the-art. Any new components should be inter-operarable with existing components. The Consultant will then provide conceptual-level designs for any new or upgraded ITS.

57. **Preliminary Cost Estimate.** The Consultant will provide a preliminary cost estimate for each of the corridors, including utilities, footpaths, cycle ways, road construction, bus stops, landscaping, street lighting, signage, road markings, and ITS. Cost estimates should be developed using prices obtained from similar contracts awarded in Armenia or in the region, procured on an international basis, cross checked with similar contracts awarded in the region.

58. **Deliverables:**

(i) Full conceptual-level designs for the bus priority corridors, stations, off-corridor stops, staging areas, and points along the off-corridor routes, utilities, and ITS, including costing; and

(ii) 3-D renderings of bus stations, with two different top structure architectural designs.

Task 3: Traffic and Parking Analysis

59. As the repurposing of general traffic lanes for bus priority lanes may impact general traffic, it will be necessary to conduct a traffic analysis. Additionally, as the development of a bus priority corridor may impact parking supply, it will be necessary to evaluate options for displaced parking. This task should be undertaken simultaneously and in an iterative manner with the conceptual design for the bus priority corridors (Task 2). Data collection may be done in tandem with the data collection required in Task 1.

60. **Road traffic crash data:** The Consultant will obtain and analyze road crash data on the bus priority corridors and affected bus routes (see Task 1) to establish a baseline for road safety including fatal and serious injury crashes disaggregated by gender, age, and mode.

61. **Traffic and Parking Surveys.** The Consultant will conduct the following traffic surveys:

(i) **Midblock counts:** Hourly traffic and pedestrian counts midblock throughout the bus priority corridors, in the peak hour. If traffic congestion during the peak hour is so high

that traffic volumes appear suppressed during that time, the Consultant should also collect traffic volumes at the highest volume time in which traffic is moving. This may be in the hour before or after the peak.

(ii) **Intersection counts:** Peak hour intersection classified turning counts at a selection of intersections, prioritized based on the level of complexity of the potential design improvements at each intersection, and agreed upon with the PIU. Pedestrians should be included in these counts.

(iii) **Parking Supply Survey:** The Consultant shall document all on-street parking spaces and curb regulations along the bus priority corridors, and for approximately 100m on every cross street. The Consultant shall also document all off-street parking spaces within a 100m radius of the bus priority corridors. The Consultant should produce parking supply maps which indicate parking spaces and curb regulations. Where parking fees exist, the team should note any parking fee levels and include them on maps.

(iv) **Parking Occupancy Survey:** To estimate parking demand, the Consultant should lead its team to record the number of parked vehicles at each hour in each of the above-documented spaces, classified by vehicle type, as well as duration, in order to estimate turnover rates. The Consultant should produce parking demand maps which indicate average turnover rates for each block.

62. **Traffic Analysis:** The Consultant shall analyze traffic, including level of service and queuing under the proposed conceptual design. As part of this work, the Consultant shall document expected improvements in mobility, congestion reduction, and overall traffic flow resulting from the implementation of the bus priority corridors.

63. **Traffic Management Plan:** The Consultant will prepare a traffic management plan for the corridor, including the development of recommended possible remedial actions that can be taken to mitigate negative impacts on residents and merchants. Routing maps for mixed traffic should be prepared, showing how existing trips can be made, under potentially new traffic conditions and signaling.

64. **Microsimulation:** The Consultant will select five to ten key locations of high traffic volumes and develop microsimulations at those locations both with and without the bus priority infrastructure, as designed in Task 2.

65. **Parking analysis:** Due to the inclusion of new bus lanes along the corridors, some loss of on-street parking may be necessary. Using the parking supply and occupancy data collected above, the Consultant will assess the degree to which parking loss can be mitigated through displacement to nearby locations. This will require a careful evaluation of parking demand on the corridor and off the corridor, as well as the possibility of creating more supply off-corridor where, for instance, parking spaces may not be well-delineated.

66. **Deliverables:**

(i) Traffic and parking surveys and analysis report.
(ii) Traffic Management Plan, including microsimulation.
(iii) Parking analysis report

Task 4: Conceptual Design and Upgrading for 2 Trolleybus Depots

67. In this task, the Consultant will develop the conceptual designs to rehabilitate two trolleybus depots which are currently in poor condition. The depots should be retrofitted to house a total of 249 18-meter battery electric buses.

- Bagratunyats Depot
- Jrvezh Depot

68. The Bagratunyats Trolleybus Depot on Bagratunyats St near Arshakunyats Avenue is a minimally serviced bus depot operated by Yerevan Bus. It is approximately 42,000 m2. This is sufficient space to accommodate a minimum of 146 18-meter buses, 215 12-meter buses, or 316 8-meter buses.



Figure 3: Bagratuniats Bus Depot

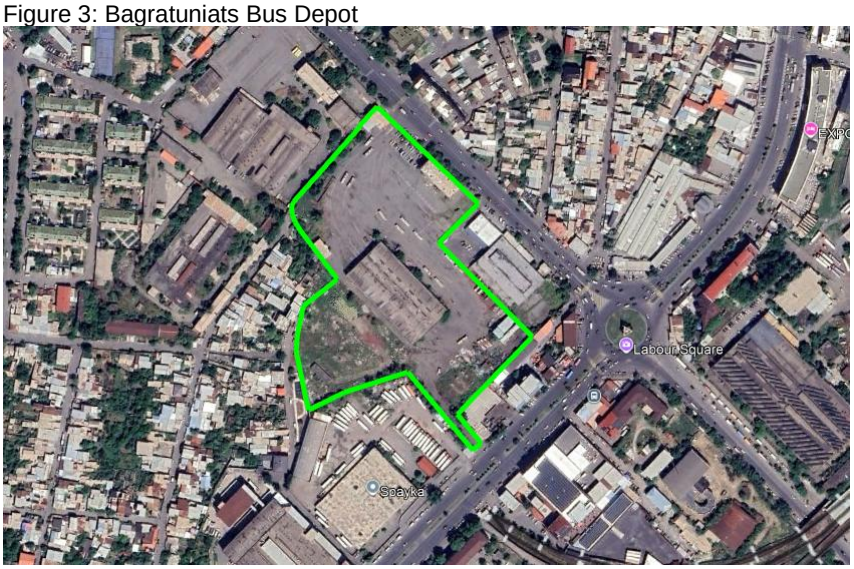


Figure 4: Aerial view of Bagratunyats Depot

69. The (Nor Nork) depot Tevosyan Street. It is approximately 33,000 m2. It currently houses an estimated 2/3 of the 250 active fleet. It also houses an additional 200 parked Euro3 buses that were donated in the past but currently are not being used. It also currently houses Yerevan's CNG bus fleet. This depot currently has diesel fuel supply.



Figure 5: *Existing Jrvzeh Trolleybus Depot*

70. **Review and assess existing depot conditions**. Existing site plans for both depots can be found in Annex A. The Consultant will review these and conduct site visits to both depots to confirm measurements and conditions of the existing facilities. The Consultant will assess the depot structures to determine whether any portion of the structures at either depot site can be retained or whether the entire structure should be rebuilt. Assessment of utilities at the depot site will also be needed. The Consultant will also review the existing power capacity available at each depot. Additionally, the Consultant will review any plans for depot renovation or upgrades. Finally, the Consultant should assess the number of electric buses which are possible to house in each depot, considering the possibility that some buses currently housed in each depot may need to be housed in other depots currently in existence.

71. **General layout and circulation plan.** For both depots, the Consultant will develop a circulation plan, with associated drawings, to inform the conceptual design of the depot and associated facilities. If any portion of any structure is to remain on site, the location of that structure should be included in the circulation plan. If, however, any structures need to be fully rebuilt, the Consultant may consider whether a change in location or structure size would make sense. For this, the location of utilities will also need to be considered.

72. The plan should include an overall circulation of vehicles throughout the depots, along a main circulation pathway, as well as circulation within each area of the depots.

73. The general flow of buses within the depot should aim to be as follows, space and existing structures notwithstanding: Vehicles will enter through a secure gate. An RFID (Radio Frequency Identification) connection will log when the vehicle has entered. First, vehicles will be inspected at the inspection area, where they will be visually checked to determine whether they will need to go to maintenance. If so, they would go directly to the maintenance area. Otherwise, they have the option of recharging. To do this, they would travel to the recharging station. Next, buses would travel to a wash area, which should be indoors. In most cases, they would then either travel to a light maintenance area for routine maintenance, or to a parking area. If a serious issue gets

flagged at the light maintenance area, or on the road, buses will go to the heavy maintenance area. The Consultant may also elect to combine the heavy and light maintenance areas into a single larger area. Buses that are finished with maintenance will park until their next dispatch. At that point, they will exit the depot through a separate gate from the one at which they entered.

74. Staff will enter the depot through an entrance separate from the buses, either by car, motorcycle, bicycle, or foot. Cars, motorcycles, and bicycles will travel directly to the staff parking lot where there will be adequate parking spaces. The staff vehicle exit gate will be adjacent to the entrance gate.

75. There should be an assigned parking spot for each bus based on its registration or serial number, to simplify bus operations inside the depot and ensure efficiency in receiving the bus from handing it over to the assigned crew. A fixed or assigned bay is also essential for providing on-site servicing of buses, such as charging of buses, cleaning, and air pressure monitoring. Fixed bays inside the bus parking area ensure optimum parking layout of buses, avoiding stacking and loss of usable space due to inefficient parking. One-way circulation within the depot parking, coupled with 60-degree angled parking, allows maximum efficiency in terms of space use and ease of operations (pulling in and out of the bays).

76. Both depots will include an administrative facility. In the administrative facility, functional areas should be organized around a central corridor to provide convenient access. This will simultaneously allow for the independence, and controlled access (to ensure safety and security) of each function. Rest facilities, toilets, and the cafeteria should be accessible from both exterior and interior entrances while offices and test and training facilities should be accessible from the interior only.

77. **Conceptual design of the depots.** Depot design will include but is not limited to, the following sub-components:

- Entry/exit area. Entrances should be provided at an edge of the major access road leading to the depot. For buses, an entry gate with a check post including an RFID connection for in-shedding, should be provided. A separate exit, also equipped with an RFID connection for out-shedding, should be provided to optimize bus flow. For staff and visitors, a single entry/exit gate, separate from the buses, should be provided to access the administration block and parking. A dedicated room for security personnel should be stationed at the bus entry / exit area.

- Charging station and battery swapping facilities. Vehicle charging stations should be placed in the overnight parking area. Fast chargers should also be included. The Consultant should confirm the number of needed chargers and placement of the PV panels. Air inflation points are also suggested to be installed in the charging station area, two at each charger.

- Vehicle washing area. External washing of buses should be mechanized, and include all surfaces i.e., sides, top and bottom of the bus. A fully mechanized washing system using gantry washers should be considered, as well as water recycling systems. An enclosed washing area should be specified. Under-body cleaning should be undertaken manually through hose pipes. The consultant should confirm the number of needed wash bays, though typically, one bay is recommended for every 72-75

buses. The Consultant will conduct a detailed analysis of the bus depot layout to determine the optimal location for the vehicle washing area. Design a layout that allows for easy access and maneuverability of buses during the washing process. Consider the flow of vehicles, water supply, and wastewater disposal in the layout.

Include a water recycling system to collect, treat, and reuse wastewater to minimize water wastage during the washing process. Integrate a separate water supply system for non-potable uses, such as vehicle washing, to conserve potable water.

Include a wastewater collection and treatment system to efficiently capture wastewater from the washing area and to prevent pollution and comply with environmental regulations. Plan and ensure the safe disposal of treated wastewater, or recycling of treated wastewater adhering to environmental regulations. Design waste collection points and recycling facilities within the washing area.

Design the vehicle washing area to accommodate potential expansion of the bus fleet in the future.

- Bus parking area. The bus parking area should be at-grade and should include designated bays for each bus. The angle of the parking should be 60-degree angle parking, wherever possible. The parking area should be sheltered by the PV panels. The parking area should also include plug-in chargers.

- Inspection and Maintenance area: Confirm the number of pits needed in the inspection and maintenance area. The pits shall be accessed by fixed stairways as well as by a single ramp which leads to all pits, so that they are interconnected. Inspection and maintenance facilities should be designed specifically for electric buses, taking into account the particular needs for battery lifecycle management. A drainage system shall be included for the pits.

- Maintenance supervisor offices. At least two offices should be provided in proximity to the maintenance area for supervisory staff. These offices should be enclosed with some protection from noise but with visual sightlines of the maintenance area. Water closets (toilets, wash basins, etc.) shall be provided in general proximity to the maintenance and workshop area.

- Workshops. There should be separate workshops for various components and activities. There should be one for engines, one for gears, one for painting (which requires air-conditioning and filtering), one for upholstery, one for tire resolving, and one for electrical equipment. Parts for each of these components should be stored in these workshops.

- Parts storage room. One main parts storage room with separate sections for old and new tires, oil tanks, all other spare parts, and scraps.

- Tools storage room. A tools storage room shall be provided to systematically organize, store, and secure work tools. The arrangement of the tools' storage room shall be such that regular accounting and auditing of the tools can be easily facilitated.

- Administrative offices. A reception and security area shall be provided on the ground floor of the administrative building. Public conference room areas may also be located on this floor. The administrative offices should be provided on the floors above the ground floor. At least three (3) executive-sized rooms shall be provided. At least fifteen (15) technical staff office rooms shall be provided. Open workstation areas shall also be provided for administrative staff. Conference room facilities shall also be provided. The office area shall also include a copy machine room, office supplies storage, and a pantry and light lunch area. The administrative office area should be equipped with dedicated storage areas for fire and emergency equipment, first aid supplies, and janitorial supplies. Quality and adequately sized male and female water closets (toilets, wash basins, etc.) shall be provided on each floor of the administrative building; at least one toilet in each water closet shall be disability accessible.

- Driver and conductor facilities. Quality male and female locker rooms shall be provided for drivers, conductors, and other bus company staff. The number of lockers provided shall be sufficient for at least two full shifts of drivers and conductors, relative to the number of buses in each depot. Quality male and female water closets (toilets, basins, etc.) shall be provided, including provision of disability-accessible toilets. A cafeteria and kitchen shall also be provided. The cafeteria should be readily accessible to both driver and conductor crews as well as administrative staff. Rest and recreational facilities shall also be provided for the staff, including a television/board games room and some form of physical exercise area (e.g., table tennis, basketball, etc.). The crew locker rooms, rest/recreational areas, water closets, and canteen should be provided near the entrance of the administrative block with access from outside the depot. A room for staff training, which may also serve as a conference room, should be provided and a driver simulation machine should be considered and costed. A laundry room in proximity to the locker rooms shall be provided to facilitate uniform washing.

- Staff parking. This should be outside of the operational area and should include enough parking spaces for staff cars, motorcycles, and bicycles, as well as disabled parking near the building entrances.

- Utilities. The Consultant will provide conceptual designs for all utilities which connect into the various facilities being designed in the depots. The Consultant must prioritize energy efficiency and sustainability throughout the utility design. This approach aligns with green building practices, incorporating renewable energy sources and environmentally friendly materials.

- Landscaping. The depot and the adjacent bus stop should be attractive from a landscaping perspective.

- Perimeter fencing. The depots should be secured with fencing around the perimeter with minimal visual impact. For the walls near residential homes, acoustic panels should be utilized to minimize noise. Likewise, a green wall (i.e., wall with creeper plants) or a wall with an outside line of trees should be specified so that any visual impact on the community is minimized.

78. Each complex should be barrier-free and accessible to all, including persons with disabilities. At the entrances to the site and other buildings in the complex, ramps should be provided to accommodate level changes for wheelchair users. Elevators should be installed in the buildings with floors/functions at higher levels.

79. Both depots will need to provide power for plug-in electric vehicles. The Consultant will assess the capacity and condition of existing power supply substations, and recommend rehabilitation, modernization and upgrading as appropriate.

80. A solar photovoltaic (PV) system holds the potential to reduce operational costs. Solar panels should be placed over the parking area of the new depots, providing an additional power supply for the fleet, as well as weather protection. The Consultant will analyze the possibility of putting the collected power back into the municipal power grid rather than, or in addition to, storing the power on the premises of the depot. The Consultant will review the current capacity of the electric company to receive distributed power into the grid at the depots, its feasibility and reliability, and the terms under which such power input into the grid is compensated. The Consultant will make a recommendation with respect to whether it is more economically optimal and operationally secure to return the power to the grid or store the collected power on the premises of the depot. The Consultant will then issue a report providing the pros and cons of each approach, including the financial, economic and operational ramifications of each alternative.

81. The Consultant will then specify the number and type of PV panels required for each depot, as well as the amount and type of power storage equipment, and the required connections to the bus charging stations. They will specify the estimated lifecycle costs and the likely power the equipment can be expected to reliably supply.

82. Deliverables:

(i) Report reviewing existing conditions at depots and future plans

(ii) Conceptual design for both depots.

Task 5: Communications and Branding

83. **Public Awareness Campaign:** Undertaking new bus priority corridors requires a significant amount of community outreach and public buy-in. Customers' bus routes will be changed, and that must be handled with care. There is also the potential to bring new customers to the bus system if routes are designed in an attractive way. The Consultant team will undertake a public awareness campaign.

84. Using the materials developed in previous tasks, the Consultant will develop a project website which includes the high-quality renderings, all the maps, information about route changes, and a project timeline. The website will be well-designed, easy to use, and updated regularly.

85. The Consultant will lead at least three public workshops to present the project and will provide handouts. The Consultant will take note of any major concerns and will discuss them with the appropriate parties, to be clarified in the inception stage. If, as a result of any major concerns, said parties request changes to the project, the Consultant will consider those changes on their technical merits and, where appropriate, will reflect any changes in all materials, as well as in the service plan.

86. **Branding:** In order to generate excitement around the bus priority corridors, the Consultant will develop a cohesive brand for the system. The Consultant will develop a list of names for the system and will present them to the PIU and relevant stakeholders. Once a name has been agreed on, the Consultant will develop a logo for the system which will be used on all materials going forward. The branding designer will also design a totem to be used at all high-profile bus stops and stations.

87. **Stakeholder Engagement Plan for the Project.** The Consultant will prepare this plan summarizing the requirements during project implementation for communication to stakeholders, consultations, and grievance redress mechanisms, as presented in other project documents.

88. **Deliverables**:

(i) PowerPoints, handouts, and minutes from three public workshops

(ii) Website for the project

(iii) System brand including name and logo

(iv) Stakeholder engagement plan

Task 6: Safeguards

89. The Consultant will support the EA and implementing agency (IA) in preparing and reviewing all required safeguard documents in accordance with ADB's Safeguard Policy Statement 2009 (SPS). The Consultant will adhere to the guidance in ADB's SPS and should follow the Environment Health and Safety Guidelines, the Good Practice Note on Addressing Sexual Exploitation, Abuse, and Harassment in ADB-Financed Projects with Civil Works, meaningful consultation and grievance and redress mechanism (GRM).

90. **Environmental Safeguards.** The Consultant shall:

(i) support the EA/IA in preparing initial environmental examination (IEE) or environmental impact assessment (EIA) depending on project categorization, including an environmental management plan (EMP) in accordance with relevant policies and

guidelines of the Government of Armenia and ADB, and ADB's Safeguard Policy Statement *2009* and its guidelines;

(ii) provide guidance to and collaborate with counterpart national consultant to set up the baseline information for monitoring and conduct additional surveys if applicable;

(iii) assist the EA/IA in organizing meaningful public consultations at the project sites, inviting local stakeholders, and affected persons, and documenting the findings of the consultations. In the case of environment category A projects, carry out at least two meaningful public consultations (once during the early stage, and once when the draft EIA report is available). Develop a stakeholder involvement plan and safeguards grievance redress mechanism for project implementation as part of the project's EMP. Translate the final draft EMP into Armenian;

(iv) assess the capacity of the EA/IA to plan, manage, implement, and monitor safeguard activities, and prepare capacity-building measures and training workshops as needed.

(v) facilitate effective coordination between ADB and the EA/IA, and other stakeholders on environmental safeguard aspects of the project.

91. **Resettlement Safeguards.** The Consultant shall:

(i) assess the capacity of the EA/IA in preparing and implementing land acquisition resettlement plans (LARPs), including monitoring; specify clear institutional set-up and responsibilities in order for each group to function effectively and coordinate efficiently; design and implement capacity-building interventions (such as workshops and training) during LARP preparation; which should be reflected in the LARP;

(ii) assist the EA/IA in preparing the project's LARP, following SPS 2009;

(iii) conduct meaningful consultations and prepare a consultation and disclosure strategy as part of the LARP that specifically includes stakeholder mapping, tools (focus group discussions, group meetings, etc.), and consultation and disclosure materials; the section on consultation in the LARP should include the summary of consultation and disclosure activities carried out during planning, issues and concerns raised by the affected persons and other stakeholders, and a matrix on how feedback was used or incorporated in the LARP; the LARP should include all minutes of meetings as an attachment;

(iv) assess existing grievance redress process of the government and agree with the government on a localized GRM with clear responsibilities, reporting requirements and budget necessary for the committee to function effectively; carry out the required training specific to handling grievances and document them in the resettlement plans;

(v) facilitate effective coordination between ADB and the EA/IA, relevant government agencies, and other stakeholders on resettlement safeguard aspects of the project.

92. **Deliverables:** The Consultant shall produce the following deliverables:

(i) IEE or EIA, including EMP.

(ii) LARP.

Task 7: Economic and Financial Analysis of the Project

93. The Consultant will complete economic and financial due diligence in compliance with ADB's requirements.

94. **Financial Management Assessment.** The Consultant will be responsible for developing the required Financial Management Assessment and Financial Analysis document for the ensuing loan. The consultant will perform, but not limited to the following tasks:

(i) Provide capacity strengthening to the EA/IA on ADB procedures for financial analysis and financial management assessment.

(ii) Conduct a financial management assessment in accordance with ADB guidelines of the EA and IAs, including the following:

a. assessing whether previous financial management assessments have been conducted by ADB or other agencies and, if so, reviewing the results and ascertaining whether these can be used as input

b. assess FM staffing and capacity for planning and budgeting, management and financial accounting, reporting, auditing, internal controls, and information systems,

c. review and recommend proposed disbursement and funds-flow arrangements following ADB procedures and based on project needs and the financial management assessment of the EA/IA,

d. identify risks arising from material deficiencies, and propose practical recommendations for improving the financial management functions and/or staffing for financial operations needed to mitigate risks,

(iii) identify financial management integrity risks and mitigation measures[5] to ensure compliance with ADB's Anticorruption Policy,[6]

e. provide relevant detailed information to enable completion of the financial management sections, and

[5] Guidance available in: ADB. 2023. *Integrity Risks and Red Flags in Transport Projects*. Manila
[6] ADB. 1998. *Anticorruption Policy*. Manila.

f. explain to the government the financial management assessment, the relevant proposed financial risk mitigation measures and discuss and agree if those actions can be practically implemented.

(iv) Assess and reach agreement on financial reporting, auditing, and public disclosure arrangements for the project, and, as appropriate, identify and agree on arrangements for receiving financial statements from EA/IA.

95. **Financial Analysis.** The Consultant will be responsible for developing the required Financial Analysis Linked Document for the ensuing loan. The consultant will perform, but not limited to the following tasks:

(i) provide support in the preparation and agreement of cost estimates and a financing plan based on verifiable data and are sufficient to support project implementation.

(ii) Prepare financial projections and assess the financial performance of the EA/IA for the past 5 years and evaluate their financial capacity and incremental recurrent costs to determine financial sustainability and review proposed cost-recovery and tariff policies, including affordability.

(iii) Conduct financial evaluations (financial cost-benefit analysis) including sensitivity analysis of project components that have a cost-recovery objective.

(iv) Where significant risks are identified to project financial sustainability or viability, propose relevant financial performance indicators to be incorporated in financial covenants.

(v) Summarize the results of the financial model in a report meeting ADB's requirement for financial analysis, and perform sensitivity and risk analysis in accordance with ADB's Guidelines.

(vi) Assist the ADB project team in finalizing (i) the financial analysis; (ii) project cost tables; and (iii) financial projections.

96. **Economic Analysis.** The Consultant will be responsible for developing the required economic models for analysis, and the Economic Analysis Linked Document for the ensuing loan. The consultant will perform, but not limited to the following tasks:

(i) Conduct the economic analysis of the overall project in accordance with ADB's Guidelines for Economic Analysis of Projects and other relevant policies and guidelines.

(ii) Identify all project costs and benefits through a comparison of with- and without-project situations. Estimate the economic internal rate of return on the basis of non-incremental and incremental economic benefits and economic costs (including economic capital, operation, and maintenance costs), in constant economic prices. The economic evaluation will include environmental and poverty reduction benefits envisaged under the Project.

(iii) Undertake sensitivity analysis to assess the effects of adverse changes in key assumptions that underlie the economic analysis and financial analysis including, but not limited to project costs, traffic, implementation delay, varying tariff rates, and combination of these factors.

(iv) Assist the project team to carry out project appraisal to specify assurances and other conditions attached to the loan.

(v) Assist the ADB project team in finalizing (i) the economic analysis; (ii) project related loan covenants; (iii) assessment of the project benefits, impacts, assumptions, and risks; and (iv) design and monitoring framework, including output and outcome indicators with their respective baselines, following ADB's Guidelines for Preparing a Design and Monitoring Framework.[7]

97. **Deliverables:** The Consultant shall produce the following deliverables:

(i) Financial Management Assessment Report.

(ii) Financial Analysis Report and Linked Document.

(iii) Economic Analysis Report and Linked Document.

Task 8: Procurement Planning

98. The Consultant will undertake procurement planning activities following ADB guidelines, including:

(i) assess procurement capacity of the EA/IA.

(ii) identify and assess risks relevant to procurement and propose mitigation measures.

(iii) identify procurement integrity risks and mitigation measures[8] to ensure compliance with ADB's Anticorruption Policy,[9]

(iv) Prepare the Strategic Procurement Plan (SPP), ensuring proper contract methods and packaging in accordance with ADB's Procurement Policy, Procurement Regulations, Staff Instructions, and relevant Guidance Notes.

(v) investigate local procurement procedures from other related project procurement experience and assess the challenges and setbacks of procurement.

(vi) prepare draft bidding documents and terms of reference for various contract packages, including procurement of buses and recruitment of consultants for detailed engineering

[7] ADB. 2020. *Guidelines for Preparing a Design and Monitoring Framework*. Manila
[8] Guidance available in: ADB. 2023. *Integrity Risks and Red Flags in Transport Projects*. Manila
[9] ADB. 1998. *Anticorruption Policy*. Manila.

design of bus corridors and depots, and consultants to develop the Operational Design and Business Model.

(vii) provide procurement policy guidance as well as technical advice on strategic procurement activities to relevant officials, to ensure procurement arrangements and schedules are consistent with project implementation and development objectives.

(viii) conduct training to the EA/IA on ADB's procurement and consultant's rules, procedures, integrity requirements, and policies.

99. **Deliverables:** The Consultant shall produce the following deliverables:

(i) SPP.

(ii) Draft bidding documents and terms of reference.

Task 9: Climate Change Assessment

100. The Consultant will assess the project's climate risks, adaptation and mitigation options and climate financing following ADB's guidelines. Activities will include:

(i) Conduct a Climate Risk and Adaptation assessment (CRA) in line with international best practice
- Assess the change and variability of key climate-related parameters over the project lifetime including historic trends and projections based on latest climate models.
- Assess the risks posed by climate change to project viability, as well as the vulnerabilities of the system as a whole.
- Evaluate and recommend costed adaptation measures to incorporate into the project design and/or operation of the project.

(ii) Complete the Climate Change Assessment following the latest ADB template
- Summarize findings of the CRA providing the climate change context of the project and costed adaptation measures recommended.
- Assess additional climate change mitigation measures and provide costing and GHG reduction potential calculation.
- Estimate mitigation and adaptation climate financing following approved joint methodology and common principles Joint Methodology for Tracking: Climate Change Adaptation Finance | Asian Development Bank (adb.org) Common Principles for Climate Mitigation Finance Tracking | Asian Development Bank (adb.org)
- With data inputs from wider consultant team calculate the project GHG emissions and relative emissions following Guidelines for Estimating Greenhouse Gas Emissions of Asian Development Bank Projects: Additional Guidance for Transport Projects | Asian Development Bank (adb.org)
- Finalize the Paris Alignment Assessment Guidance Note on Implementing Operations' Alignment with the Paris Agreement at ADB and its addenda for intermediated financing and policy-based lending.

101. **Deliverables:** The Consultant shall produce the following deliverables:

(i) Climate Risk and Adaptation Assessment (CRA)

(ii) Climate Change Assessment.

Task 10: Poverty, Social and Gender

102. The Consultant will undertake poverty, social and gender assessments of the project following ADB guidelines. Activities will include:

(i) Review the current poverty reduction policies and regulations in the project areas.

(ii) Identify the main population groups that may be affected by the project (positively/negatively and directly/indirectly) and their social characteristics, and conduct follow up interviews/consultations with relevant subgroups that may be affected differently (women, ethnic minorities, people with disabilities and other vulnerable groups).

(iii) Design the household survey methodology, including sampling and questionnaire, use survey results to assess the social, gender, and poverty impact of the project, and conduct a distribution analysis.

(iv) Survey project beneficiaries by gender, ethnicity, and income groups; identify vulnerable groups including female-headed households, migrant workers and people with disabilities, and estimate beneficiaries falling below the official poverty line.

(v) Prepare a consultation and participation plan and conduct key informant, focus group discussions and other types of qualitative interviews with all relevant stakeholders, e.g., civil society organizations, public transport workers, private sector firms, relevant government agencies, etc.

(vi) Assist with the formulation of contingent valuation survey in consultation with the project economist and conduct affordability analysis for improved services including low-income households, and the beneficiaries' willingness to pay the tariffs.

103. Conduct a socio-economic impact analysis in accordance with ADB's Handbook on Poverty and Social Analysis (2012), Technical Note on Social Analysis of Transport Projects and Sustainable Transport Initiative and prepare socioeconomic and poverty profiles for areas affected by the project; working with other team members identify and confirm key social and poverty issues and options to address these issues including

assessment of various risks such as sexual exploitation abuse and harassment (SEAH)[10], HIV/AIDS, trafficking, and labor issues.

(vii) Establish a consultation and participatory process including stakeholders' analysis for the Project that will continue throughout the project design, planning, implementation to ensure stakeholders are identified and their needs and demands are considered.

(viii) Conduct a gender analysis, including SEAH and assess the degree to which the project could bring about gender benefits, with an aim to develop the project as 'gender equity theme' category in accordance with ADB guidelines.

(ix) Assess the institutional and social issues affecting participation including that of civil society organizations (CSO), ownership, acceptance and adoption of project activities and propose strategies to overcome any identified issues.

(x) Prepare and develop social development and gender action plan (SDGAP) which will address other social dimensions (i.e., the poor, gender, disabled, elderly, etc.) as well as non-safeguard issues (i.e., labor, health, safety, etc.) and design an appropriate consultation and participation plan for the project implementation. Ensure that the SDGAP is formulated in coordination with the EA/IA, discussed with them and endorsed by the EA/IA.

104. **Deliverables:** The Consultant shall produce the following deliverables:

(i) Poverty, Social and Gender Assessment Report

(ii) Social Development and Gender Action Plan

Task 11: Institutional Assessment and Capacity Building

105. The objective of this task is to build the capacity of the government in Yerevan to manage this, and other related projects.

106. **Institutional Analysis:** The Consultant shall map the current organizational framework for transport management in Yerevan and determine any clear gaps in efficiency. This will require describing, in detail, the existing structure and mandates of each agency involved in managing transport in Yerevan, developing organigrams. The Consultant will answer questions related to the management of each department – who does the head of the department report to, how are they appointed, etc.

107. The Consultant shall then identify challenges with the current structures which relate to the delivery of the types of tasks listed in this TOR. These could include problems

[10] ADB. 2023. *Good Practice Note on Addressing Sexual Exploitation, Abuse, and Harassment in ADB-Financed Projects with Civil Works*. Manila.

of duplicate mandates, capacity gaps, revenue stream complexities, asset management, vehicle maintenance, etc.

108. The Consultant shall then identify specific staff members whose work relates to transport planning and operations in Yerevan and map their roles and responsibilities.

109. Based on the analysis conducted above, the Consultant shall identify the institution and staff members in the best position to manage contracts with companies who would carry out the tasks described in this TOR.

110. **Institutional and Capacity Development Road Map**: Based on the institutional assessment conducted above, a road map for institutional capacity development will be prepared, following ADB-CDIA guidelines, covering institutional, organizational, and human resource development. This includes a training needs assessment (TNA).

The road map will have short, medium, and long-term components. Short-term recommendations form an action plan, while medium-term components support a dedicated capacity program.

111. **Knowledge Transfer:** The consultants will transfer technical knowledge through on-the-job training and knowledge sharing. A report on these initiatives will be submitted.

112. **Deliverables:** The Consultant shall produce the following deliverables:

(i) Institutional assessment

(ii) Institutional and capacity building roadmap

(iii) Report on knowledge transfer activities

6. Reporting Requirements and Time Schedule for Deliverables

113. **Duration.** Consulting services under the proposed contract will be implemented over a maximum period of **12 months** (March 2025 through February 2026).

114. **Deliverables.** The Consultant will prepare milestone reports (main text of maximum 100 pages, with unlimited annexes attached) as summarized in Table 2 below and will submit 5 copies to ADB-CDIA and the EA and the PIU, together with an electronic version. Comments received from the review panel (para. 121 below) should be promptly addressed and incorporated by the Consultant within one week of receipt. Detailed contents of the reports and deliverable schedule will be agreed between ADB-CDIA, the EA/IA, the IPIU, and the Consultant at inception stage.

Table 2: Summary of Milestone Reports

Milestone Report/Event	**Content**	**Expected Delivery Date***	
Mobilization & Inception Report *followed by Inception Workshop*	- First project assessment, incl. review of data availability, objectives, schedules, and outputs; - Confirmation of scope, approach/ methodology; - Detailed work plan covering all activities.	**CS + 2 weeks**	
Service Plan for the Bus Priority Corridors	**Output 1:** Memo describing selection of considered routes for further analysis **Output 2:** Data collection methodology memo **Output 3:** Raw and processed datasets for all collected data (Excel), including maps **Output 4:** Network diagnostics PPT **Output 5:** Service Plan PowerPoint and report, including high-quality, graphic-designed maps of the network, each route, and changes to existing routes.	**CS + 4 months**	
Conceptual Design for Two Bus Priority Corridors	**Output 6:** Full conceptual-level designs for the bus priority corridors, bus stations, off-corridor stops, utilities, ITS, points along the off-corridor routes, and costings **Output 7**: 3-D renderings of stations, with three different top structure architectural designs	**CS + 6 months**	
Traffic and Parking Analysis	**Output 8:** Traffic and parking surveys and analysis report. **Output 9:** Traffic Management Plan, including microsimulation. **Output 10:** Parking analysis report	**CS + 6 months**	

Conceptual Design and Upgrading for 2 Trolleybus Depots	**Output 11:** Report reviewing existing conditions at depots and future plans **Output 12:** Conceptual design for both depots.	**CS + 7 months**	
Communications and Branding	**Output 13:** PPTs, handouts, and minutes from three public workshops **Output 14:** Website for the project **Output 15**: System brand including name and logo **Output 16**: Stakeholder engagement plan	**CS + 7 months**	
Safeguards	**Output 17:** IEE/EIA **Output 18**: LARP	**CS + 8 months**	
Economic and Financial Analysis of the Project	**Output 19**: Financial Management Assessment Report **Output 20**: Financial Analysis Report and Linked Document **Output 21**: Economic Analysis Report and Linked Document	**CS + 8 months**	
Procurement Planning	**Output 22:** Strategic Procurement Plan **Output 23:** Terms of reference	**CS + 8 months**	
Climate Change Assessment	**Output 24:** Climate Risk and Adaptation Assessment **Output 25:** Climate Change Assessment	**CS + 8 months**	
Poverty, Social and Gender	**Output 26:** Poverty, Social and Gender Assessment Report **Output 27:** Social Development and Gender Action Plan	**CS + 8 months**	
Institutional Assessment and Capacity Building	**Output 28:** Institutional Assessment **Output 29**: Institutional and Capacity Building Roadmap	**CS + 8 months**	

	Output 30: Report on Knowledge Transfer Activities		

Note: * Months after commencement of services (CS)

115. **Workshops.** The Consultant will organize and conduct meetings with the EA, PIU, and ADB-CDIA as needed, as well as several workshops and thematic working sessions at specific milestones, to guarantee mutual understanding of project issues and proposed options.

116. All expenses for those workshops should be borne by the Consultant under their consulting contract. The PIU and Consultant can jointly determine whether the workshops are face-to-face or through video teleconferencing.

117. **Reporting requirements.** In addition to main deliverables, the Consultant will prepare monthly progress reports (main text of maximum 10 pages), summarizing:

(i) Main technical issues encountered during the past month and major technical decisions to be taken urgently;

(ii) Administrative issues such as (a) financial and physical progress of the consultancy contract; (b) arrivals and departures of each team member; (c) elapsed time and person days on the assignment in the preceding month and forecast for the coming month; (d) expenditures, billing and payments received; and (e) consultants proposed changes in personnel schedule and team composition.

(iii) Minutes of important meetings with project stakeholders.

(iv) Main challenges ahead and main recommendations from the Consultant.

118. At the end of the assignment, the Consultant will also provide a summary final report including comparison of actual expenditures compared to budget, consultant inputs compared to work plan, and other assessments which could be useful in designing future assignments of a similar type.

119. **Donor Visibility.** The ADB-CDIA branding guidelines will be shared upon the issuance of the Notice-to-Proceed (NTP). The consulting firm will be responsible for assisting the City of Yerevan in ensuring the appropriate visibility of donor resources. All reports and documents generated under this assignment must explicitly acknowledge ADB-CDIA's support and display the logos of their respective donors. Additionally, the consultant is required to recognize these donors in any public communications related to the project.

120. **Language requirements.** All reports will be presented in English language. If required by the EA, a translation in Armenian language of main reports may be provided. In such case, the Consultant will ensure that documents in Armenian will have the same title as the English version and will follow the same page layout as the English version. Consistency and integration of proposed changes (track change mode) will be ensured.

121. **Procedure for review and acceptance of project outputs.** Before formal acceptance, each of the outputs will be peer reviewed by a review panel composed of:

(i) The IA

(ii) The Head of the PIU; and

(iii) The ADB project Team Leader.

122. At each workshop held after reaching each milestone, the review panel will jointly:

(i) Determine if all key deliverables were provided as agreed; and

(ii) Review and discuss the quality of the outputs submitted in each report.

123. The Consultant will be given, within 10 working days, a written list of comments or deficiencies to be addressed and the requested date of revised submission. The Consultant will be given the opportunity to clarify the nature and extent of the deficiencies and agree with ADB and the government on the needed revisions and resubmission date. The new submission will then be reviewed again by ADB and the government to determine whether the deficiencies have been sufficiently addressed, in which case the report/output will be deemed as "final," and the progress payment will be approved.

7. Team Composition and Qualification Requirements for the Key Experts and Non-key Experts

124. **Team Leader.** The Consultant shall appoint a Team Leader to be responsible for the overall management of the work described in these terms of reference, for contact with the EA/IA and ADB, providing work plans and progress reports, coordinating the inputs of the individual experts and for all official dealings with the EA. The Team Leader shall have power of attorney to act on behalf of the Consultant in any contractual matters regarding the assignment.

125. **Team composition and required inputs:** A total of 70.7 person-months (pm) will be required to complete the proposed assignment including: (i) 42.7 pm of international experts; and (ii) 28.0 pm of national experts. The team should include the following key experts and non-key experts. Key experts will be evaluated according to the criteria defined in the request for proposals. Non-key experts will be evaluated on a pass or fail basis.

126. **International key experts:**

i) **Team Leader/Senior Bus Priority Expert (international)**. The Team Leader/Senior Bus Priority Expert will be responsible for all study deliverables and will carry out, but will not be limited to, the following tasks:

   a. Coordinate and supervise inputs from all team members.

   b. Develop the conceptual design and costs for both bus priority corridors

c. Review all deliverables

d. Capacity building and knowledge transfer reporting

e. Reporting

ii) **Senior Bus Operations Specialist (international)**. The Senior Bus Operations Specialist will carry out the following tasks:

a. Lead oversight over data collection and data processing.

b. Manage the Graphic Designer to create visuals of the bus-related data.

c. Identify affected bus routes

d. Lead the network diagnostics by evaluating the considered routes and identifying issues.

e. Develop bus priority corridor service plan and generate fleet, demand, and station sizing estimates

f. Identify staging locations

g. Conduct workshops on the bus priority corridor service plan

h. On-the-job capacity building

iii) **Senior Bus Priority Station Architect (international)**. The Senior Bus Priority Station Architect will carry out the following tasks:

a. Development of concept architectural design for Bus Priority stations

b. Work with graphic designer to develop station renderings

c. Development of concept architectural design for bus depots

iv) **Senior Traffic Management Specialist (international)**. The Senior Traffic Management Specialist will carry out the following tasks:

a. Oversee traffic surveys

b. Traffic impact analysis

c. Development of intersection designs and analysis

d. Traffic signal timings and specifications

e. Microsimulation of traffic flow at 5-10 locations with and without the proposed bus priority project

f. Document cause of delay at bottleneck locations and propose interventions

g. On-the-job capacity building

v) **Senior NMT Design Specialist (international)**. The Senior NMT Design Specialist will carry out the following tasks:

a. Conceptual design of NMT access to bus priority stations

b. Conceptual design of NMT portion of bus priority corridors

c. On-the-job capacity building

vi) **Senior Bus Depot Design Specialist (international).** The Senior Bus Depot Design Specialist will carry out the following tasks:

a. Assessment of two existing bus depots

b. Circulation plan for rehabilitation of bus depots

c. Conceptual design of the bus depots, including upgrading for battery electric buses.

127. **International non-key key experts:**

vii) **Senior ITS Specialist (international)**. The Senior ITS Specialist will carry out the following tasks:

a. Assessment of ITS needs for the bus priority system
b. Conceptual design, description, and costing of ITS for the bus priority system

viii) **Senior Parking Management Specialist (international)**. The Senior Parking Management Specialist will carry out the following tasks:

a. Collect parking data along and near the bus priority corridors
b. Parking management strategy

ix) **Senior Communications and Branding Specialist (international)**. The Senior Communications and Branding Specialist will carry out the following tasks:

a. PPTs, handouts, and minutes from three public workshops
b. Project website
c. System brand including name and logo
d. Stakeholder engagement plan

x) **Senior Utilities Specialist (international)**. The Senior Utilities Specialist will carry out the following tasks:

a. Assessment of existing utilities in the bus priority corridors
b. Conceptual design of utilities in the bus priority corridors

xi) **Environment Safeguards Specialist (international).** The Environment Safeguard Specialist will carry out the following tasks:

a. Environmental due diligence

b. IEE

xii) **Social Development Specialist (international).** The Social Development Specialist will carry out the following tasks:

a. Poverty, Social and Gender Assessment (data disaggregated)
b. Social Development and Gender Action Plan

xiii) **Transport Economist (international).** The Transport Economist will carry out the following tasks:

a. Economic analysis
b. Economic Analysis Report and Linked Document

xiv) **Climate Change Specialist (international).** The Climate Change Specialist will carry out the following tasks:

a. Climate Risk and Adaptation Assessment
b. Climate Change Assessment

xv) **Procurement Specialist (international).** The Procurement Specialist will carry out the following tasks:

a. Procurement Planning
b. Draft bidding documents/Terms of reference

xvi) **Senior Institutional Specialist (international).** The Senior Institutional Specialist will carry out the following tasks:

a. Conduct Institutional Analysis
b. Institutional and Capacity Building Roadmap
c. Oversee capacity building and report on knowledge transfer activities.

128. **National key experts:**

i) **Deputy Team Leader / Urban Transport Specialist (national)**. The Deputy Team Leader / Urban Transport Specialist will carry out the following tasks:

a. Manage the team of national consultants
b. Coordination with the international team
c. Coordination with local government
d. Provide inputs into all work products

129. **National non-key experts:**

ii) **Communications Specialist (national)**. The Communications Specialist will carry out the following tasks:

a. Develop a directory of stakeholders
b. Organize stakeholder engagement
c. Develop materials for stakeholder engagement
d. Assist with website development

iii) **Data Collection Manager (national).** The Data Collection Manager will carry out the following tasks:

a. Oversee survey team during data collection

b. Data entry post-survey

iv) **Environment Safeguards Specialist (international).** The Environment Safeguard Specialist will carry out the following tasks:

c. Environmental due diligence

d. IEE

v) **Social Specialist (national)**. The Social Specialist will carry out the following tasks:

e. Due diligence for all social aspects of the project including the Poverty and Social Analysis

f. Ethnic Groups Due Diligence Report

vi) **Gender Specialist (national)**. The Gender Specialist will carry out the following tasks:

g. Due diligence for all gender-related aspects of the project

vii) **Resettlement Specialist (national)**. The Resettlement Specialist will carry out the following tasks:

a. Due diligence for all physical activities and potential land acquisition and resettlement

b. Due diligence for impacts for businesses due to loss of access

c. Preparation of the LARP

viii) **Cost Specialist (national)**. The Cost Specialist will carry out the following tasks:

a. Provide local inputs for cost estimation

ix) **Graphic Designer (national)**. The Graphic Designer will carry out the following tasks:

a. Graphic design for reports

b. Bus priority corridors and station high-quality 3-D renderings

c. 3-D renderings for bus depots

d. Graphic design of public outreach materials

e. Graphic design for logo and branding

x) **Financial Analysis and Management Specialist (national).** The Financial Analysis and Management Specialist will carry out the following tasks:

a. Financial Management Assessment Report

b. Financial Analysis Linked Document

xi) **Procurement Specialist (national).** The Procurement Specialist will carry out the following tasks:

a. Procurement due diligence activities. Support in preparation of SPP, draft bidding documents/Terms of reference

130. The required qualifications and experience of the international experts are shown in **Table 3**.

Table 3: International experts

	Position	Person-months	Experience requirements
1	Team Leader, Bus Priority Expert	6.0	The Team Leader/Bus Priority Expert will possess at least a bachelor's degree in civil engineering, transport engineering, transport / urban planning, project management, or a related field, and preferably the expert will also possess a post-graduate degree. The expert will possess a minimum of 15 years of relevant experience in implementing sustainable urban transport projects, including a minimum of 8 years in designing and implementing bus priority and BRT systems. The expert will also have at least 10 years of experience in a position managing large sustainable urban transport projects as the Team Leader.
2	Senior Bus Operations Specialist	3.6	The Senior Bus Operations Specialist will possess a university degree in transport / urban planning or a related field. The expert will possess at least 10 years of relevant experience. The Senior Bus Operations Specialist should have conducted surveys for bus/bus priority operational design and developed service plans specifically for bus priority systems in at least three projects, and should have experience in the management of battery-electric bus fleets.
3	Senior Bus Priority Station Architect	3.2	The Senior Bus Priority Station Architect will possess a university degree in architecture, engineering, or a related field. The expert will possess at least 10 years of relevant experience in designing public transport stations, and experience designing Bus Priority stations in at least three bus priority or BRT systems.
4	Senior Traffic Management Specialist	6.0	The Senior Traffic Management Specialist will possess a university degree in civil engineering or a related field. The expert will possess at least 10 years of

	Position	Person-months	Experience requirements
			relevant experience in designing traffic surveys, traffic analysis, and microsimulations, with a similar role in at least two bus priority/BRT projects.
5	Senior NMT Design Specialist	2.4	The Senior NMT Design Specialist will possess a university degree in transport/urban planning or a related field. The expert will possess at least 10 years of relevant experience in designing NMT facilities (bicycle and pedestrian) and at least one project in which they integrated NMT facilities into a bus priority or BRT project.
6	Senior Utilities Specialist	2.3	The Senior Utilities Specialist will possess a university degree in civil engineering or a related field. The expert will possess at least 10 years in the design of utilities for street/corridor redesign projects.
7	Senior Bus Depot Design Specialist	3.2	The Senior Bus Depot Design Specialist will possess a university degree in civil engineering or a related field. The expert will possess at least 10 years of relevant experience in designing quality urban public transport systems, with specific experience designing at least 2 bus depots.
8	Senior ITS Specialist	1.3	The Senior ITS Specialist will possess a university degree in computer science, transport, civil engineering or a related field. The expert will possess at least 10 years of relevant experience in designing ITS systems for bus-based public transport.
9	Senior Parking Management Specialist	1.6	The Parking Management Specialist will possess a university degree in transport / urban planning, engineering, or a related field. The expert will possess at least 10 years of relevant experience, with at least 3 projects in which they surveyed parking supply/occupancy and analyzed the parking situation in a city.
10	Senior Communications and Branding Specialist	3.6	The Communications and Branding Specialist will possess a university degree in communications, marketing, or a related field. The expert will possess at least 10 years of relevant experience in providing communications and branding for urban public transport systems.
11	Environmental Safeguards Specialist	2.0	The Environmental Safeguards Specialist will possess a university degree in environmental assessment, environmental management, urban planning, engineering, or a related field. The expert will possess at least 8 years of relevant experience on preparation of EIA / IEE studies for similar projects, monitoring and implementation of environmental safeguards, including specific experience with projects financed through international financial institutions.

	Position	Person-months	Experience requirements
12	Social Development Specialist	1.5	The Social Development Specialist will possess a university degree in social science, development studies, or a related field. The expert will possess at least 8 years of relevant experience in the monitoring and implementation of social safeguards, including specific experience with projects financed through international financial institutions.
13	Transport Economist	1.5	The Transport Economist will possess a university degree in urban planning, economics, or a related field. The expert will possess at least 10 years of relevant experience in developing economic analyses for transport projects financed through international financial institutions.
14	Climate Change Specialist	1.0	The Climate Change Specialist will possess a university degree in urban planning, environmental science, or a related field. The expert will possess at least 10 years of relevant experience in preparation of Climate Risk and Adaptation Assessments for similar projects, including specific experience with projects financed through international financial institutions.
15	Procurement Specialist	2.0	The Procurement Specialist will possess a university degree in business, finance, accounting, law, engineering, or a related field. The expert will possess at least 10 years of relevant experience in managing procurement for development projects.
16	Senior Institutional Specialist	1.5	The Senior Institutional Specialist will possess a university degree in business, public policy, international development, or a related field. The expert will possess at least 10 years of relevant experience in mapping institutions, particularly in the transport sector, for development projects.

131. The required qualifications and experience of the national experts are shown in **Table 4**.

Table 4: National Experts

	Position	Person-Months	Experience requirements
1	Deputy Team Leader / Urban Transport Specialist	6.0	The Deputy Team Leader/Urban Transport Specialist will possess a university degree in project management, business, engineering, transport / urban planning, or a related field. The expert will possess at least 10 years of relevant experience in developing and managing urban transport projects.
2	Data Collection Manager	2.3	The Data Collection Manager will possess a university degree in civil engineering. The expert will possess at least 7 years of relevant experience in managing surveys for transport planning purposes.

	Position	Person-Months	Experience requirements
3	Communications Specialist	3.2	The Communications Specialist will possess a university degree in communications, media relations, journalism, business, or a related field. The expert will possess at least 7 years of relevant experience in providing communications and stakeholder engagement assistance to projects.
4	Environmental Safeguards Specialist	1.5	The Environmental Safeguards Specialist will possess a university degree in environmental assessment, environmental management, urban planning, engineering, or a related field. The expert will possess at least 5 years of relevant experience on preparation of EIA / IEE studies for similar projects, monitoring and implementation of environmental safeguards, including specific experience with projects financed through international financial institutions.
5	Social Specialist	1.5	The Social Specialist will possess a university degree in social science, development studies, or a related field. The expert will possess at least 5 years of relevant experience in the monitoring and implementation of social safeguards, including specific experience with projects financed through international financial institutions.
6	Gender Specialist	1.0	The Gender Specialist will possess a university degree in social science, gender studies, development studies, or a related field. The expert will possess at least 5 years of relevant experience on the monitoring and implementation of gender safeguards, including specific experience with projects financed through international financial institutions.
7	Resettlement Specialist	2.5	The Resettlement Specialist will possess a university degree in social science, development studies, or a related field. The expert will possess at least 7 years of relevant experience in the monitoring and implementation of resettlement safeguards, including specific experience with projects financed through international financial institutions.
8	Financial Analysis and Management Specialist	1.5	The Financial Analysis and Management Specialist will possess a university degree in urban planning, economics, or a related field. The expert will possess at least 7 years of relevant experience in developing financial management assessment and financial analysis for transport projects financed through international financial institutions.
9	Procurement Specialist	1.5	The Procurement Specialist will possess a university degree in business, finance, accounting, law, engineering, or a related field. The expert will possess at least 7 years of relevant experience in managing procurement for development projects.

	Position	Person-Months	Experience requirements
10	Cost Specialist	1.5	The Cost Specialist will possess a university degree in civil engineering or a related field. The expert will possess at least 7 years of relevant experience in costing of major infrastructure projects, contributing local costs inputs.
11	Graphic Designer	5.5	The Graphic Designer will possess a university degree in graphic design, urban planning, or a related field. The expert will possess at least 5 years of relevant experience with graphic design, at least 3 of those years focusing on urban projects.

132. **Site office and facilities.** The Consultant shall establish its workplace in Yerevan in an office to be arranged by the firm. The Consultant shall include, in its estimates, provisions for potential rent of an office and for all necessary supplies and equipment, IT hardware and software, including printers necessary for the performance of the work.

133. **Software and Licenses.** Where the Consultant provides software (except for ordinary MS-office programs) in order to conduct their duties, e.g. provision of a planning tool, the software (or the license) including all data in them shall be the property of the EA. Software shall be provided in a sufficient number of copies to conduct the work properly. The Consultant shall provide the necessary licenses for the software for the work to be conducted properly by the EA (for at least 2 users – experts and counterpart) and the Licenses shall be without time limitations if such licenses are available from the supplier. If not, the licenses shall be valid for one year.

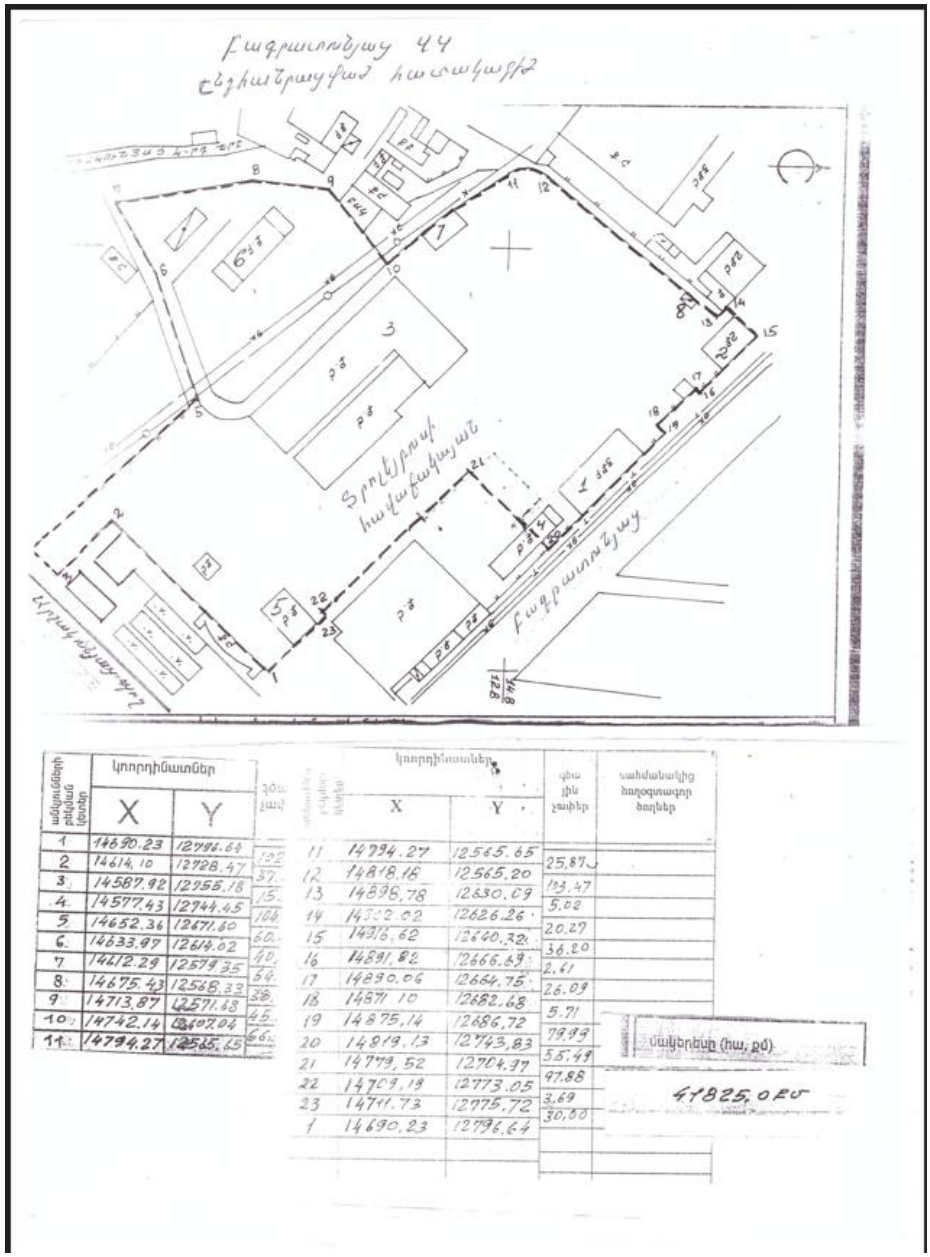
134. **Living accommodation.** The Consultant must provide living accommodations for all team members while working on the assignment and include the out-of-pocket expenses in their proposal.

135. **Transport.** The Consultants must provide their own transport for work and other activities while working on the assignment and include the out-of-pocket expenses in their proposal.

8. Client's Input and Counterpart Personnel

136. The Client will provide technical counterparts, as required, to form part of the Team, and will provide the Consultant with all important available data and information.

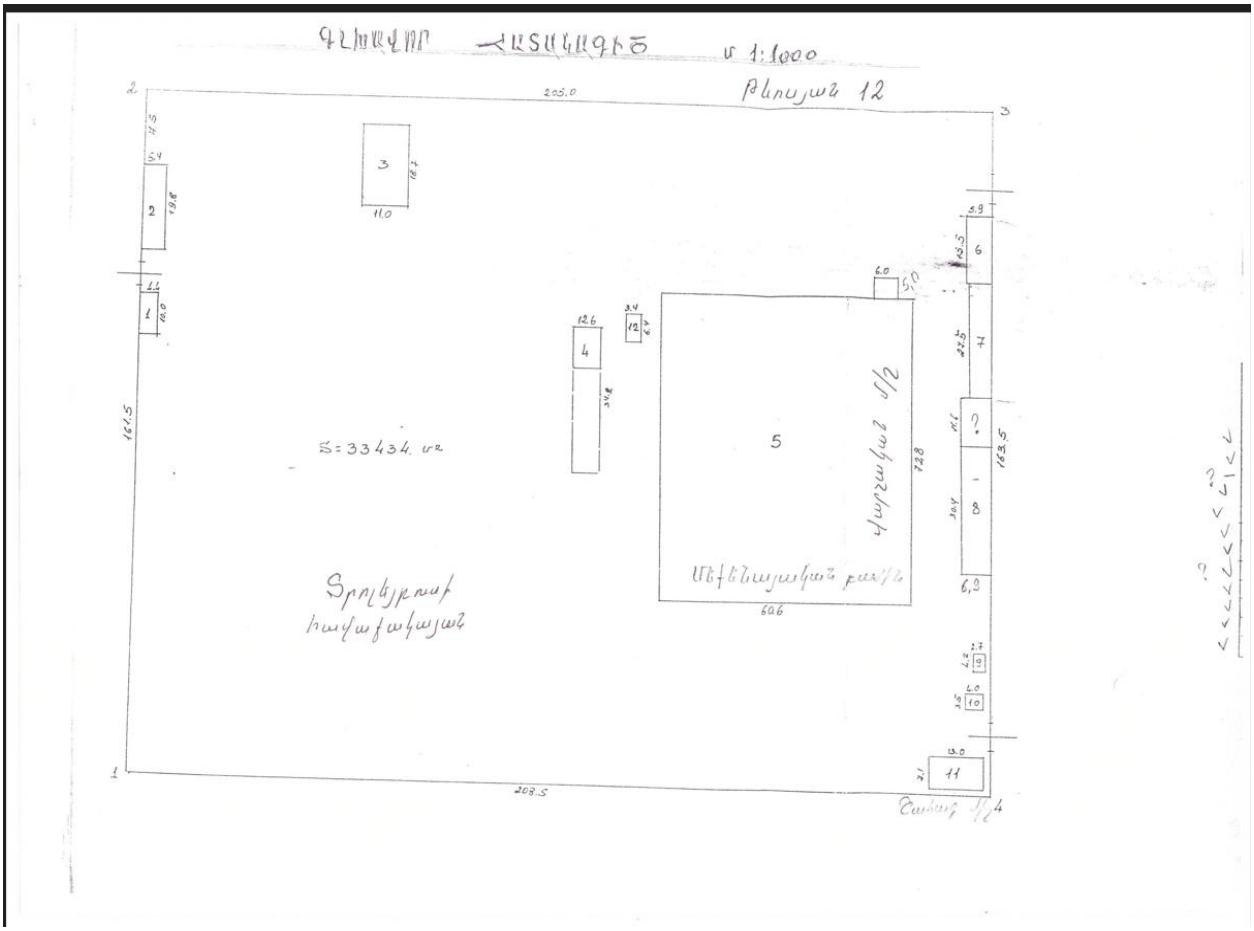
ANNEX A: EXISTING DEPOT SITE PLANS



անկյունների թվանշանային նշումները	կոորդինատներ X	Y	չափ
1	14690.23	12798.64	
2	14614.10	12728.47	103.47
3	14587.92	12755.18	37.
4	14577.43	12744.45	15.
5	14652.36	12671.60	104.
6	14633.97	12614.62	60.
7	14672.29	12579.35	52.
8	14675.43	12568.33	11.
9	14713.87	12571.68	38.
10	14742.14	12597.04	45.
11	14794.27	12565.65	60.
12	14818.16	12565.20	25.87
13	14896.78	12630.09	103.47
14	14908.02	12626.26	5.02
15	14916.62	12640.32	20.27
16	14891.82	12666.69	36.20
17	14890.06	12664.75	2.41
18	14871.10	12682.68	26.09
19	14875.14	12686.72	5.71
20	14919.13	12743.83	79.99
21	14779.52	12704.97	55.49
22	14709.19	12773.05	97.88
23	14711.73	12775.72	3.69
1	14690.23	12798.64	30.80

Սակերեսը (հա, քմ)

41825.0 քմ


1:1000
Բևռայա 12
205.0
208.5
161.5
183.5
S=33434. մ2
Տրոլեյբուսի
հանգույցային
Վերելակային բակ
6.0
11.0
18.7
13.8
12.6
3.4
3.4
5.4
4.5
5.9
6.9
2.7
4.0
13.0
2.1